USER TESTING REPORT

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Evaluating Designs with Users

An Analysis of JetBlue.com

March 15, 2023

EXECUTIVE SUMMARY

In this study, we researched the question, “Can frequent travelers use JetBlue.com to plan their trips?” and determined participants’ success in accomplishing specified tasks that are realistic and common to our targeted user population. Participants in this study aligned with the recruiting criteria of having booked at least one flight online in the past year and having never used the site before. The User Tests were conducted one-on-one where each participant was observed while facilitating the assigned tasks. We were able to determine key points of interest that should be acknowledged to improve the usability of the JetBlue flight booking website.

Results

Although there were some critical moments where participants had some issues navigating the website efficiently, they were all able to complete the tasks successfully. Additionally, the participants rated their experience and understanding of the website as a high usability ranking according to the SUS Post-Test Questionnaire with an average score in the 91.5 percentile. The critical moments appeared in the tasks of booking a one-way flight, finding the menu option for sharing the chosen flight itinerary, and booking a flight when the user is open to flexible dates.

Recommendations

It is recommended that you use a pop-up notification to ask the user if they are interested in booking a one-way flight since this is often the only reason a user leaves the “return flight” field blank.

Additionally, when a user views their cart, it is recommended that the designers replace the 3-dot menu with icons to provide the user with a clear view of all options available to them.

Also, it is recommended that the team replace the word “remove” with a trash can icon to decrease the cognitive load on users and prevent errors.

Last, it is recommended that you place a checkbox near the date fields so users who want to book a flight with flexible dates can search without needing to located the “Best Fare Finder” tool within the menu.

INTRODUCTION

In this study, we are evaluating Jet Blue's flight booking system on JetBlue.com. This system is used by a wide variety of individuals who are interested in booking a flight on JetBlue Airlines. Our team is observing a sample population of users to provide recommendations for designing a system with great usability, reduced frustration, and a high success rate—thus accomplishing the goal of increasing sales.

With the overarching question, "Can frequent travelers use JetBlue.com to plan their trips?" we are collecting data through interviews, surveys, and usability testing.

METHODS

Target Population

The target population for this study is the general customer of JetBlue Airlines. We want to know if the average potential customer can successfully navigate the website producing a high completion rate, average to low task timings, and low error rate.

The recruitment criteria for this study includes two parameters:

1. The participant must have purchased at least one airline ticket online in the past year.
2. The participant must not have used JetBlue.com previously.

Recruitment

To recruit individuals for this study, we used our personal connections. All individuals in this study are people that we know personally to fit the recruitment criteria.

Summary of Participants

Participant 1

- 21 years old
- Lives in Ann Arbor, Michigan
- Booked 1 flight online in the past year

- Proficient in technology
- Graduate student at the University of Michigan

Participant 2

- 60 years old
- Lives in Ann Arbor, Michigan
- Booked 1 flight online in the past year
- Proficient in technology
- Works as an engineering consultant

Participant 3

- 43 years old
- Lives in St. Petersburg, Florida
- Booked 3 flights online in the past year
- Proficient in technology
- Works as a teacher

Participant 4

- 32 years old
- Lives in St. Petersburg, Florida
- Booked 12 flights online in the past year
- Proficient in technology
- Works as a teacher

Participant 5

- 27 years old
- Lives in Saline, Michigan
- Booked 3 flights online in the past year
- Proficient in technology
- Works as a computer programmer

Tools/Instruments

Script

User Test Script (built upon User Test Script provided in Coursera Course "Evaluating Designs with Users)

Introduction to User Test Demonstration

Hi, thanks for being here today! As you know, we are trying to improve the JetBlue website, and getting your honest feedback will be a great help to us. The goal for today's session is test out the JetBlue website. I'm here to see how you navigate through the website.

User Test Script

I'll start this session by asking some background questions, and then have you complete a set of tasks. As you work on the tasks, please think aloud by telling us every thought that crosses your mind as you navigate the website. Don't be afraid to be wrong or unsure. In this test, mistakes are actually a good thing because it will help our developers know what to focus on. Tell us what you're attempting to do and how you think it should work. Tell me if you feel confused or if something works well. Keep in mind that if you do get stuck, I want to first see you can try to figure it out on your own. Don't worry about timing. You can have as much time as you need.

What questions do you have before we begin?

Consent Form

Present Consent form, explain the synopsis, and obtain signature.

Pre-test Questionnaire

1. Have you used _____ before?
2. Tell me about the last flight you booked.
    1. What airline did you fly?
    2. What platform did you use to book your trip?
    3. Who were you booking a flight for?
    4. What was your reason for travelling?
3. What information is the most important to you when you are planning a flight?
4. How often do you travel?

User Test Script

Task Instructions

Read each task aloud to the participant, allowing time for the participant to complete each test. Give the printed list of tasks to the participant so they can refer back to the instructions while completing tasks.

Debrief

1. **Review of Critical Moments:**
    1. What difficulties did you have?
    2. I noticed you struggled with____, what were you thinking at the time?
    3. You paused here. What were you thinking then?
2. **Preferences:**
    1. How do you feel about the site?
    2. What did you find that you liked? What were some things you didn't like?
    3. What parts of the page were most helpful to you?
3. **Changes:**
    1. If you could choose 2 things to make the site better, what would they be? Why?
4. **Understanding:**
    1. How would you explain the process to your friend?
5. **Use Cases:**
    1. Would you use this site again? For what purpose(s)?

Conclusion

Thank you for your time and your thoughtful responses today. I am so glad you mentioned _____. We will take it into consideration as we provide feedback to the developers. We value your ideas and truly appreciate you taking the time to be here today. Thank you so much.

Questionnaire

Post-test questionnaire

Answer the following questions based on the scale of 1 -5 where 1 is strongly disagree and 5 is strongly agree

1. I think that I would like to use this system frequently

   1 2 3 4 5

2. I found the system unnecessarily complex

   1 2 3 4 5

3. I thought the system was easy to use

   1 2 3 4 5

4. I think that I would need the support of a technical person to be able to use this system

   1 2 3 4 5

5. I found the various functions in this system were well integrated

   1 2 3 4 5

6. I thought there was too much inconsistency in this system

   1 2 3 4 5

7. I would imagine that most people would learn to use this system very quickly

   1 2 3 4 5

8. I found the system very cumbersome to use

   1 2 3 4 5

9. I felt confident using the system

   1 2 3 4 5

10. I needed to learn a lot of things before I could get going with this system.

    1 2 3 4 5

Task Descriptions

Task 1

You're interested in trying out a new airline called JetBlue. You decide to book online at JetBlue.com. Plan a one-way trip for you to fly from Detroit to Boston departing May 26th , 2023. You want to book the cheapest fare. Share the itinerary by sending an email.

Task 2

You want to take a trip with your friends. Plan a round trip for 4 from Detroit to JFK May 1, 2023 to May 4, 2023. What is the cheapest total price for the trip?

Task 3

You want to fly from Detroit to New York (JFK) on June 14th and stay for a week, then fly from JFK to London Gatwick for a week before returning home. You want to fly Blue fare the whole way. What is the cheapest price for the trip?

Task 4

You live in Atlanta. You have a job interview in Boston on June 3, 2023 at noon. You want to leave on June 2 after 9am, and can arrive anytime before 6pm that night. How many flight options do you have?

Task 5

You want to use your JetBlue points to book a roundtrip flight from Phoenix (PHX) to Boston (BOS) June 10-12, 2023. You have 15,000 points. Do you have enough points to book the flight?

Task 6

You are flexible on dates, so you want to find the cheapest days to travel from Detroit to Tampa and back anytime in March or April. What dates offer the best deal?

Analysis Methods

Video review

Analysis of critical incidents

Logging sheet review

Statistical analyses

FINDINGS AND RECOMMENDATIONS

SUMMARY RESULTS

Results

Although there were some critical moments where participants had some issues navigating the website efficiently, they were all able to complete the tasks successfully. Additionally, the participants rated their experience and understanding of the website as a high usability ranking according to the SUS Post-Test Questionnaire with an average score in the 91.5 percentile. The critical moments appeared in the tasks of booking a one-way flight, finding the menu option for sharing the chosen flight itinerary, and booking a flight when the user is open to flexible dates.

Recommendations

It is recommended that you use a pop-up notification to ask the user if they are interested in booking a one-way flight since this is often the only reason a user leaves the “return flight” field blank.

Additionally, when a user views their cart, it is recommended that the designers replace the 3-dot menu with icons to provide the user with a clear view of all options available to them.

Also, it is recommended that the team replace the word “remove” with a trash can icon to decrease the cognitive load on users and prevent errors.

Last, it is recommended that you place a checkbox near the date fields so users who want to book a flight with flexible dates can search without needing to located the “Best Fare Finder” tool within the menu.

KEY FINDINGS

Key Finding 1 – Errors when booking a one-way flight

When participants were completing the first task, 4 out of 5 participants began entering the flight information into the fields for “From,” “To,” and “Depart” before clicking “Search flights.” They then received an error messaging saying “Please enter a valid date.” One participant entered the same date in both the depart and return boxes thinking this would signal to the system that they want a one-way flight. They then proceeded to choose a flight option. It was only upon seeing the second flight option screen that they realized the system was giving them a round-trip flight all on the same date. Three participants saw the “Please enter a valid date” message and realized that they forgot to tell they system they wanted a one-way flight only and were able to correct the error before moving forward. See fig. 1.

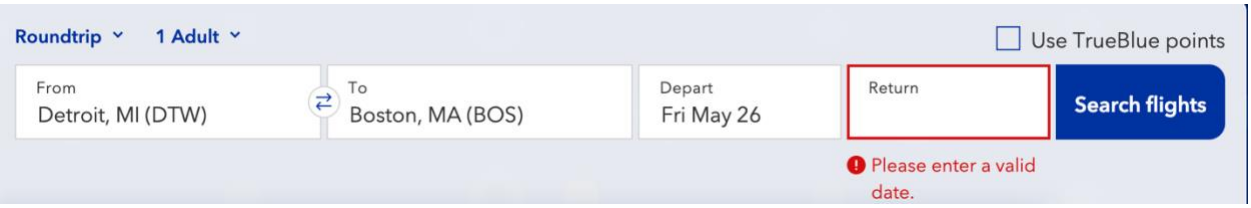


Fig 1 shows the error message a user receives when a return date is not entered.

See Recommendation #1 below.

Key Finding 2 – Need more clarity/visibility of all menu options “in the cart”

When attempting to share the itinerary, all participants first noticed the “save for later” option and clicked that. (See fig. 2) They then went back to look for a better match for the directions to share the itinerary. 1 out of 5 participants accidentally clicked “remove” when they meant to click on the 3 dot menu. 1 participant purposely clicked remove thinking it would “unsave” the flight for later. Their flights were immediately removed from the cart and they had to go back and select it again. No participants automatically knew that the “share” option would be located within the 3-dat menu.



Fig. 2 Show the options to save for later, remove, and see more options.



Fig. 3 shows the current 3-dot menu

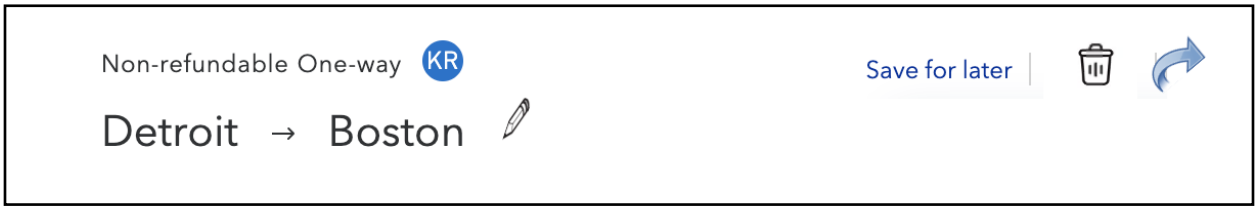


Fig. 4 shows the recommended changes to replace the 3-dot menu

See Recommendation #2, #3, #4 below.

Key Finding 3 – Best Fare Finder is challenging to locate

When trying to book a flight with flexible dates, all 5 participants first tried to leave the date fields blank. Upon receiving the error messages in red (see Fig. 5), participants looked for an alternative way to book a flight with flexible dates. 4 participants mentioned that they would expect to see a button for flexible dates near the date fields. Participant 1 clicked on "Travel Info" next. Participant 3 clicked on "Manage Trips" next, then "Book" and hovered over the "Flights" icon. They thought out loud, "I think this is going to take me back where I was." After an average of 1.92 minutes, participants were able to locate the "Best Fare Finder" options under the "Book" menu (see Fig. 6). The timing of this task is concerning. According to Jakob Nielsen, simple tasks should only take users about a minute to complete.

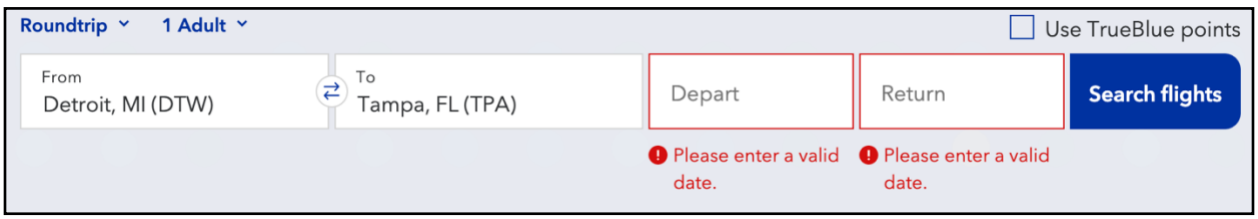


Fig. 5 shows the error messages received when date fields are left blank.

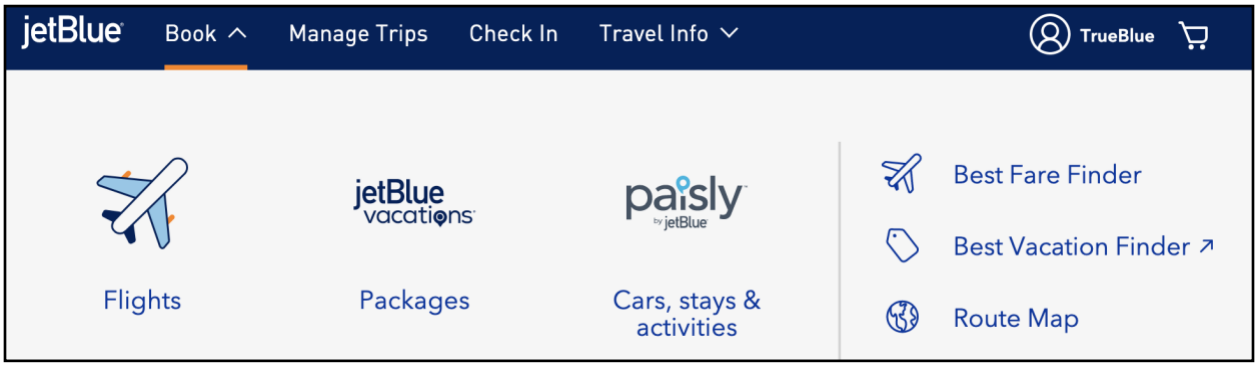


Fig. 6 Shows the “Best Fare Finder” option that was challenging for participants to find.

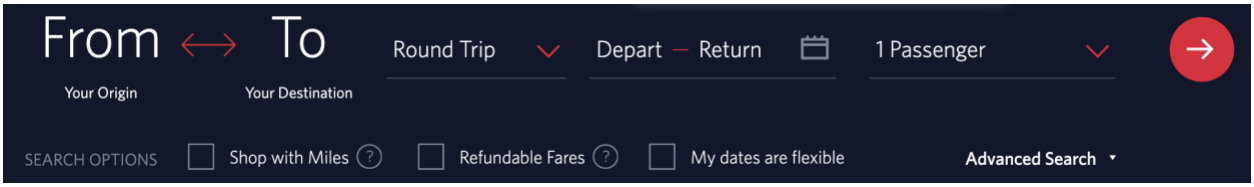


Fig. 7 Shows a competitor’s website that offers the option to check the “My dates are flexible” box upon entering the rest of their flight search information.

See Recommendation #5 below.

RECOMMENDATIONS

1. Because they system is automatically set up for booking a round-trip ticket, most often when the error of not filling in the Return box occurs, it is on purpose. Therefore, it may be useful for a popup message to ask the user “Do you want to book a one-way flight?” where they can check yes or no and the system could automatically switch from a round-trip to a one-way menu for the user.
2. Nielson’s Fifth Heuristic says that mistakes can be prevented by warning users. Because removing a flight from your cart is an action that can’t be undone, they system should warn the user of this.
3. Nielson’s Fourth Heuristic explains how following industry conventions can decrease the cognitive load on users. Although the word “remove” is clear, a user can be confused if it means remove from cart or removed from list of saved flights. A better indication that

this action will remove the flight from the user's cart would be a trashcan icon. This is more widely recognized as a complete deletion of content rather than a "removal" from one place.

4. Because the 3-dot menu seems to be out of the user's initial perception as being a likely location for the "share" feature (see Fig. 3), the items in that menu could be integrated into the system through a pencil near the name of the flight—indicating that the name can be edited—and an outwards arrow indicating an ability to share. (See Fig. 4)
5. Include a checkbox option underneath the dates fields where a user can tell the system they would like to search for a flight under the parameters of flexible date constraints to find the best price. Placing the "Best Fare Finder" in a drop-down menu leave the user wondering where to look and may lead to some users to switch to a competitor's website where searching with flexible dates is a readily-available option.

LIMITATIONS

This study was conducted with a limited number of participants who were all proficient in technology. The participant pool does not provide insight into how a person who is new to technology would access the site. Therefore, we are unsure whether a less experienced individual would be able to successfully navigate the JetBlue website to book a flight.

CONCLUSION

We received positive feedback from all participants throughout the testing process. JetBlue.com is overall, well navigated by this target population and a site that all would choose to use again. Please review our recommendations and let us know if you would like to discuss any Key Findings further.

REFERENCES

Nielsen, J. (1994) Heuristic Evaluation. In J. Nielsen. & R. L. Mack (Eds.) Usability Inspection Methods. New York, NY: John Wiley & Sons.

Nielsen, J. (2009, October 4). Powers of 10: Time Scales in user experience. World Leaders in Research-Based User Experience. Retrieved March 19, 2023, from https://www.nngroup.com/articles/powers-of-10-time-scales-in-ux/